

Number Theory

ทฤษฎีจำนวน

Number Theory (or Arithmetic)

- Dedicated to the study of integers and their properties.
- Key ideas: **Division** and **Prime numbers**.
- A branch of pure mathematics (independent of application)
- Number theory has important applications to computer science.
 - Cryptography
 - Random Number Generation
- Ancient puzzles powering modern machines

Key Topics

1. Divisibility and Modular Arithmetic
2. Integer Representations and Algorithms
 - Modular Operations: Exponentiation

Next Week

3. Primes and Greatest Common Divisors
4. Applications of Congruences
5. Cryptography

1. Divisibility and Modular Arithmetic

- When one integer is divided by a second nonzero integer, the quotient (ผลหาร) may or may not be an integer.
 - $12 / 3 =$
 - $11 / 4 =$
- Division of an integer (b) by a nonzero integer ($a \neq 0$)
 - Dividend or numerator (b) = ตัวตั้ง
 - Divider, divisor, factor, or denominator (a) = ตัวหาร

1.1 Division

$$a \mid b \quad \text{iff} \quad \exists c : ac = b \quad (a, b, c \in \mathbb{Z}, \quad a \neq 0)$$

Definition:

- If a and b are integers with $a \neq 0$, we say a divides b if there exists some integer c such that $ac = b$
 - $a \mid b$ denotes a divides b .
 - a is a divisor or factor of b
 - b is a multiple of a
 - $a \nmid b$ means a **does not** divide b .
 - There is no integer c such that $b = ac$.

Division

- Example: Are the following statements valid?
 - $4 \mid 8$
 - $-3 \mid 9$
 - $10 \mid 0$
 - $5 \mid 7$

$$a \mid b \Leftrightarrow b / a \in \mathbb{Z}$$

Some Properties of Division

- **Theorem 1:** Let a , b , and c be integers, where $a \neq 0$. Then
 - if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- Example: $3 \mid 6$ and $3 \mid 12$, then $3 \mid (6 + 12) = 3 \mid 18$

Some Properties of Division

- **Theorem 1:** Let a , b , and c be integers, where $a \neq 0$. Then
 - if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
 - if $a \mid b$ then $a \mid bc$ for all integers c ;
 - if $a \mid b$ and $b \mid c$ then $a \mid c$;

1.2 Division Algorithm/Theorem

- When an integer (a) is divided by a positive integer (d), there is a quotient (q) and a remainder (r).

$$a / d = q \text{ remainder } r$$

$$\text{Let } a \in \mathbb{Z}, d \in \mathbb{Z}^+, \quad \exists q, r : a = dq + r$$

- **Theorem 2:** Let a be an integer and d a positive integer.
 - Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.
- Also known as **Euclidean Division**.

Division Algorithm/Theorem

$$a = dq + r$$

$$a / d = q \text{ remainder } r$$

- Definition

- Dividend or numerator (a) = ตัวตั้ง
- Divider, divisor, factor or denominator (d) = ตัวหาร
- Quotient (q) = ผลหาร
- Remainder (r) = เศษเหลือ

Division Algorithm/Theorem

Example: $11 \mid 101$

Division Algorithm/Theorem

$$a = dq + r$$

$$a / d = q \text{ remainder } r$$

$$q = a \text{ \textbf{div} } d$$

$$r = a \text{ \textbf{mod} } d$$

- Example: Find the quotient and remainder.
 - $101 / 11 \rightarrow$
 - Quotient (q) $\rightarrow$
 - Remainder (r) $\rightarrow$

Division Algorithm/Theorem

- Example: Find the quotient and remainder.
 - $-13 / 4 \rightarrow$
 - Quotient (q) $\rightarrow$
 - Remainder (r) $\rightarrow$

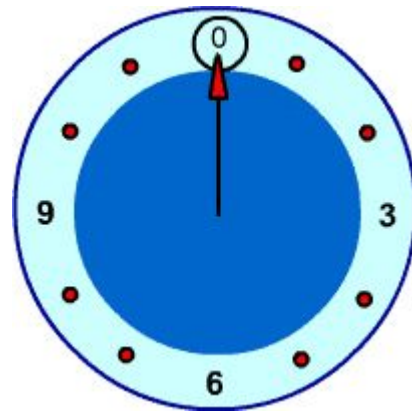
1.3 Modular Arithmetic

- In certain circumstances, we are only interested about the remainder of an integer when divided by some positive integer.
 - For these cases, there is an operator called the **modulo** operator.
 - $a \bmod d = r \quad \rightarrow \quad a \text{ modulo } d \text{ is equal to } r$
 - d is known as a modulus.
 - Example: $20 \bmod 3 = 2$

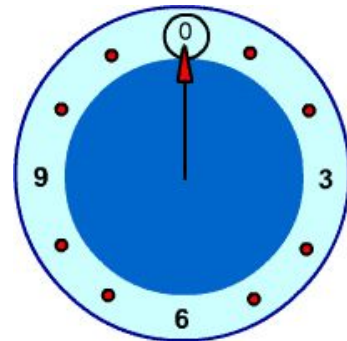
Modular Arithmetic

Real-life Example: 12-hour clock

Q: What number will be displayed on the clock if you wait 8 hours starting from now?

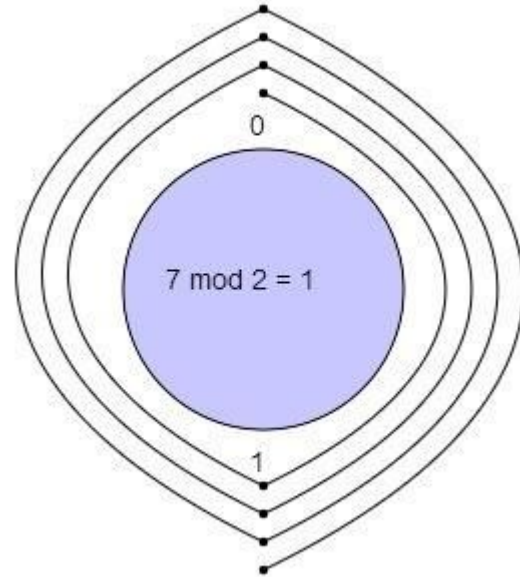
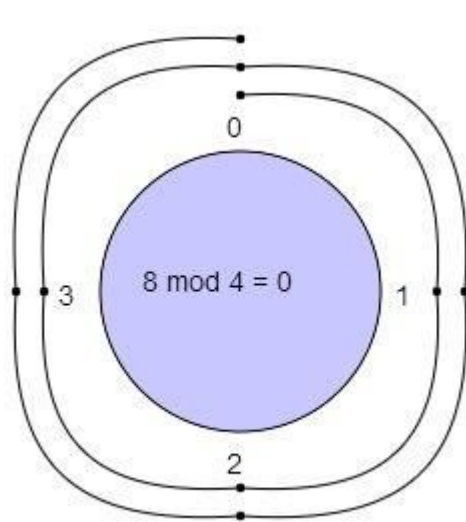
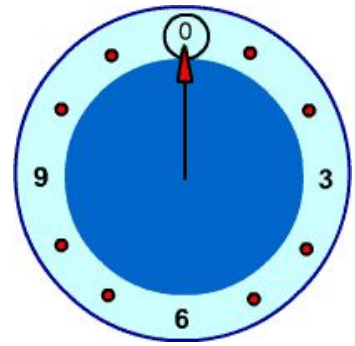


Modular Arithmetic

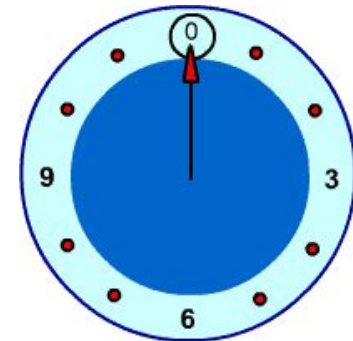
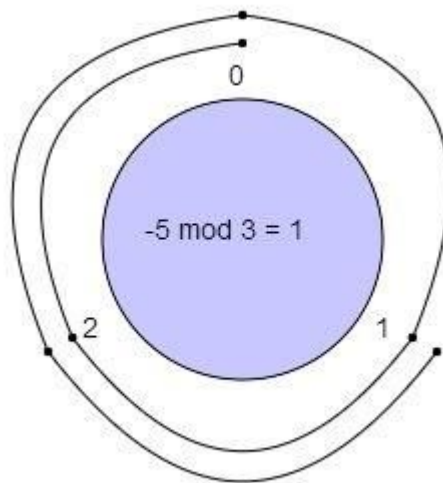


- Visualising modulo operator using circles.
 1. Write 0 at the top of a circle and continuing clockwise writing integers 1, 2, ... up to one less than the modulus (d).
 2. Start at 0 and move around the clock a steps
 - If a is positive, step clockwise
 - If a is negative, step counter-clockwise.

Modular Arithmetic



Modular Arithmetic

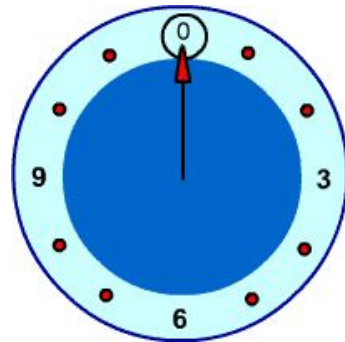


Alternatively, the formula for modulo operation when a is negative is:

$$a \bmod d = a - (\text{floor}(a/d) * d)$$

Modular Arithmetic Exercise

1. $-17 \bmod 7$
2. $-100 \bmod 8$
3. $((29 \bmod 17) \bmod 7) \bmod 3$
4. $(109 \bmod 12) + (-109 \bmod 12)$
5. $((-29 \bmod 17) \bmod 7) \bmod 3$



Congruence Modulo

If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, then

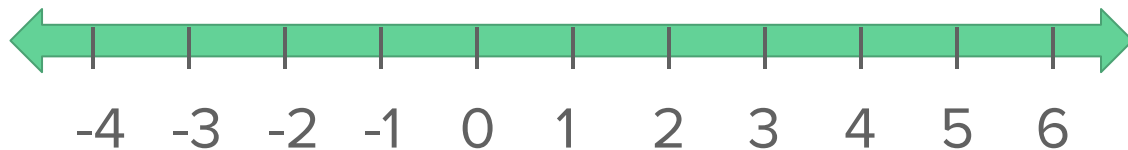
$$a \equiv b \pmod{m} \text{ if } m \mid a - b$$

- **Definition:**

- a is congruent(สมภาค) to b modulo m if m divides $a - b$.
 - Notation: $a \equiv b \pmod{m}$.

Congruence Modulo

- Example 1: Determine if $2 \equiv 6 \pmod{4}$

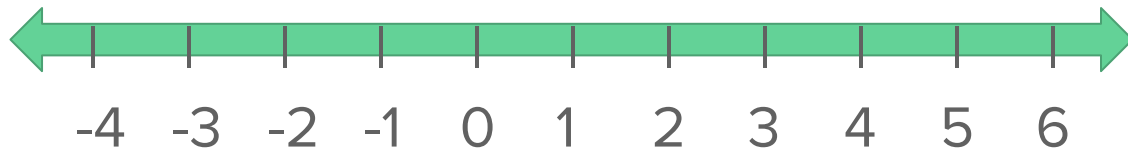


Congruence Modulo

- Example 2: Determine if $24 \equiv 14 \pmod{6}$

Modular Arithmetic

- Example: $? \bmod 4 = \{0, 1, 2, 3\}$



Congruence Modulo

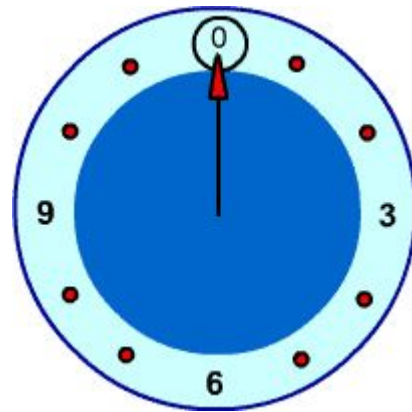
- **Theorem 3:** Let a and b be integers, and m be a positive integer.
 - Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.
 - If two integers (a and b) have the same remainder when they are divided by the positive integer m , then $a \equiv b \pmod{m}$
- Example 3: $20 \equiv 8 \pmod{12}$
 - $20 \bmod 12 = 8$
 - $8 \bmod 12 = 8$
- Note: The use of “mod” in $a \equiv b \pmod{m}$ and $a \bmod m = b$ are different.

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○ $a \equiv b \pmod{m}$ is a relation on the set of integers (congruent mod)

Congruence Modulo Example

- 12-hour clock
 - Currently it is 8:00am
 - 6 Hours later is $8 + 6 = 14$ (pm)
 - $(8+6) \bmod 12$, means 14:00 and 2:00pm
 - 14 is **congruent** to 2 modulo 12
 - $14 \equiv 2 \pmod{12}$



Congruence Modulo Exercise

- Suppose that a and b are integers,
 - $a \equiv 4 \pmod{13}$
 - $b \equiv 9 \pmod{13}$

For each statement below, find the integer c with $0 \leq c \leq 12$ such that

- $c \equiv 9a \pmod{13}$
- $c \equiv 11b \pmod{13}$
- $c \equiv a + b \pmod{13}$

Congruence Properties (Sums and Products)

- **Theorem 4:** Let m be a positive integer.
- If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
 - $a + c \equiv b + d \pmod{m}$ and
 - $ac \equiv bd \pmod{m}$

Congruence Properties (Sums and Products)

- Example 1: Find Congruence of $7+11 \pmod{5}$ and $7 * 11 \pmod{5}$

Tips: How should you use AI tools

- Never ask for a solution from start
- Practice first, then check result with AI or
- Ask for walkthrough guidance from AI
- Recommend tools
 - ChatGPT's **Study Mode**
 - Gemini's **Guided Learning**

Modular Operations

1. Addition

- $(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$

2. Multiplication

- $ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$

Modular Operations Exercise

- $(14 + 17) \bmod 5$ and $((14 \bmod 5) + (17 \bmod 5)) \bmod 5$
- $((-7) + (-10)) \bmod 7$
- $71 * 122 \bmod 11$
- $(46 * 71 * 122) \bmod 11$
- $(-45 * 77) \bmod 17$

Modular Operations

3. Exponentiation

- $a^b \bmod m = (a \bmod m)^b \bmod m$
- b can get very large, making a^b massively large and taking a long time to calculate the mod.
- For quicker calculation, divide and conquer can be used.
 - Examples: $2^{90} \bmod 13$
 - $(2^{50} * 2^{40}) \bmod 13$
 - $(2^{50} \bmod 13 * 2^{40} \bmod 13) \bmod 13 \rightarrow (4 * 3) \bmod 13$

2. Integer Representations [For AIT/DSBA]

- In the modern world, we use **decimal**, or **base 10**, notation to represent integers.
 - For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b , where b is a positive integer greater than 1.
- The bases $b = 2$ (binary), $b = 8$ (octal), and $b = 16$ (hexadecimal) are important for computing and communications.
- The ancient Mayans (2000 BC) used base 20.
- The ancient Babylonians (1895 BC – 539 BC) used base 60.

2.1 Base b Representations

Theorem 1: Let $b \in \mathbb{Z}^+$ and $b > 1$.

Then if $n \in \mathbb{Z}^+$, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a non-negative integer,

$a_0, a_1, \dots, a_k$ are non-negative integers less than b ,

and $a_k \neq 0$.

The $a_j, j = 0, \dots, k$ are called the base- b digits of the representation.

2.1 Base b Representations

Example 1: Write the decimal expansion of 12,345

$$12,345 =$$

- The representation of n given in Theorem 1 is called the **base b expansion of n** and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.

Binary to Decimal Expansion

- Most computers represent integers and do arithmetic with binary (base 2) expansions of integers.
 - In these expansions, the only digits used are 0 and 1.

Example 2: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

$$(1\ 0101\ 1111)_2 =$$

Binary to Decimal Expansion

Exercise 1: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

$$(11011)_2 =$$

Octal to Decimal Expansion

- The octal expansion (base 8) uses the digits {0, 1, 2, 3, 4, 5, 6, 7}.

Example 3: What is the decimal expansion of the number with octal expansion $(7016)_8$?

$$(7016)_8 =$$

Exercise 2: What is the decimal expansion of the number with octal expansion $(111)_8$?

$$(111)_8 =$$

Hexadecimal to Decimal Expansion

- The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols.
- The hexadecimal system uses the digits {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F}.
 - The letters A through F represent the decimal numbers 10 through 15.

Hexadecimal to Decimal Expansion

Exercise 3: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

$$(E5)_{16} =$$

Exercise 4: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

$$(2AE0B)_{16} =$$

2.2 Base Conversion Algorithm

To construct the base b expansion of an integer n :

- Divide n by b to obtain a quotient and remainder.

$$n = bq_0 + a_0 \quad 0 \leq a_0 \leq b$$

- The remainder, a_0 , is the rightmost digit in the base b expansion of n .
- Next, divide q_0 by b .

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 \leq b$$

- The remainder, a_1 , is the second digit from the right in the base b expansion of n .
- Continue by successively dividing the quotients by b , obtaining the additional base b digits as the remainder. The process terminates when the quotient is 0.

Base Conversion Example

Example 1: Find the octal expansion of $(12345)_{10}$

Base Conversion Exercise

Exercise 1: Find the binary expansion of $(241)_{10}$

Base Conversion Exercise

Exercise 2: Find the hexadecimal expansion of $(177130)_{10}$

Base Conversion

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Each octal digit corresponds to a block of 3 binary digits.

Each hexadecimal digit corresponds to a block of 4 binary digits.

So, conversion between binary, octal, and hexadecimal is easy.

Base Conversion Exercise

Exercise 3: Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.

Exercise 4: Find the binary expansions of $(765)_8$ and $(A8D)_{16}$.

Base Conversion Algorithm

Algorithm base_b_expansion(n , b : positive integers with $b > 1$)

$q = n$, $k = 0$

while ($q \neq 0$):

$a_k = q \bmod b$

$q = q \operatorname{div} b$

$k = k + 1$

return($a_{k-1}, \dots, a_1, a_0$) where $\{(a_{k-1} \dots a_1 a_0)_b$ is base b expansion of $n\}$

- q represents the quotient obtained by successive divisions by b , starting with $q = n$.
- The digits in the base b expansion are the remainders of the division given by $q \bmod b$.

2.3 Modular Operations

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2.3 Modular Operations

- Method #1: **Divide and conquer**

- $a^b \bmod c = (a \bmod c)^b \bmod c$

- If b is an even number,

- $$\begin{aligned} a^b \bmod c &= (a^2)^{b/2} \bmod c \\ &= (a^2 \bmod c)^{b/2} \bmod c \end{aligned}$$

- If b is an odd number,

- $$\begin{aligned} a^b \bmod c &= (a a^{b-1}) \bmod c \\ &= (a * (a^2)^{(b-1)/2}) \bmod c \\ &= (a * (a^2 \bmod c)^{(b-1)/2}) \bmod c \end{aligned}$$

Binary Modular Exponentiation

- In cryptography, it is important to be able to find $b^n \bmod m$ efficiently without using an excessive amount of memory, where b , n , and m are large integers.
- Method # 2: Use the **binary** expansion of n , $n = (a_{k-1}, \dots, a_1, a_0)_2$, to compute b^n .

$$b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2 + a_0} = b^{a_{k-1} \cdot 2^{k-1}} \dots b^{a_1 \cdot 2} \cdot b^{a_0}$$

- Therefore, to compute b^n , we need only compute the values of b , b^2 , $(b^2)^2 = b^4$, $(b^4)^2 = b^8$, ..., b^{2^k} and then multiply the terms b^{2^j} in this list, where $a_j = 1$.

Binary Expansion Example

Example 1: Compute 3^{11} using binary expansion of n .

Fast Modular Exponentiation Algorithm

- The algorithm successively finds $b \bmod m$, $b^2 \bmod m$, $b^4 \bmod m$, ..., $b^{2^{k-1}} \bmod m$, and multiplies together the terms $b^{2^j} \bmod m$ where $a_j = 1$.

Algorithm modular_expo(b : integer, $n = (a_{k-1}a_{k-2}\dots a_1a_0)_2$, m : positive integers)

$x = 1$

$\text{power} = b \bmod m$

for i in range(0, k):

if $a_i = 1$ **then** $x = (x * \text{power}) \bmod m$

$\text{power} = (\text{power} * \text{power}) \bmod m$

return x where $\{x \text{ equals } b^n \bmod m\}$

- $O((\log m)^2 \log n)$ bit operations are used to find $b^n \bmod m$.

Fast Modular Exponentiation Example

Example 2: Compute $3^{644} \bmod 645$ using fast modular exponentiation.

Fast Modular Exponentiation Exercise

Exercise: Compute $5^{600} \bmod 13$ using fast modular exponentiation.